"Is This Safe?" — Volunteer answer key

FOOD SAFETY

FS-K01 **SAFE** *Kids (under 10)*

You see your family washing fresh fruits and vegetables carefully under running water before eating them. They also wash their hands well before preparing the food.

Why: Washing fruits, vegetables, and hands before eating or cooking helps get rid of germs that can make people sick. When food is clean, it is safer to eat. This simple step helps keep everyone healthy.

Volunteer notes: Washing fresh produce under running water is important to remove bacteria. Good handwashing before food prep reduces germs spread. These steps help prevent foodborne illness as recommended by USDA and OSU Extension food safety basics.

Source: SP 50-1000 Turkey basics - "THOROUGHLY RINSE FRESH FRUITS AND VEGETABLES UNDER RUNNING WATER BEFORE EATING. Everyone should wash their hands before preparing food and eating."

FS-K02 **NOT SAFE** *Kids (under 10)*

You find a jar of homemade jam in the fridge with fuzzy mold growing on top. The lid has not popped up, but the jam has green and white fuzzy spots.

Why: Mold on food is not just on the surface-mold roots can go deep inside. Some molds can make you really sick even if you only see a little fuzzy part. It's safest to throw away moldy jam because it can contain poisons.

Volunteer notes: Mold often grows deep into food, producing mycotoxins that can be harmful. Visible mold means the entire food might be unsafe. USDA advises discarding moldy foods, especially soft or moist items like jams.

Source: SP 50-1000 Turkey basics - "No, you only see part of the mold on the surface. Roots invade deeply and contain poisonous substances. It's safest to discard moldy foods."

FS-K03 **IT DEPENDS** *Kids (under 10)*

You see a bowl of potato salad sitting on the kitchen table on a warm afternoon. It has been there for a little while. You wonder if it is still safe to eat.

Why: Sometimes food left out for a while can start to go bad. If the potato salad has only been out for a short time and doesn't smell funny or look strange, it might be okay. But if it's been a long time or if it smells bad or looks different, it's safer not to eat it. So, it depends on how long it's been out and how it looks and smells.

Volunteer notes: Perishable foods left at room temperature over 2 hours can grow bacteria that make people sick. The 'danger zone' is 40-140°F where bacteria grow fastest. Smell and appearance help assess safety but time and temperature history is key.

Source: SP 50-1000 Turkey basics - "If the product has been in the danger zone (40-140°F) for longer than 2-3 hours, it could be a safety concern. Watch for smells and signs of spoilage."

FS-W01 **SAFE** *Tweens (10-14)*

You helped your family put away leftover spaghetti sauce in the refrigerator right after dinner. The sauce was cooled and covered with a lid before storing it in the fridge set below 40°F.

Why: Leftovers are safe when cooled and stored properly in the refrigerator below 40°F quickly. Covering the food keeps it clean and prevents spreading germs. Eating leftovers within a few days helps ensure they are still good to eat.

Volunteer notes: Perishable cooked foods should be refrigerated promptly at 40°F or below to prevent bacteria from growing. Cover leftovers to avoid contamination and consume in a timely manner to prevent foodborne illness.

Source: EM 9331 Survival basics food - "Keep your refrigerator at 40°F or below and know when to throw food away. Refrigerate perishable food within two hours. Clean hands can save lives."

FS-W02 **NOT SAFE** *Tweens (10-14)*

You find a jar of homemade pickles in the pantry with an unsealed lid and notice the jar has tiny bubbles inside and a strange smell when opened.

Why: A jar that is not sealed properly can let harmful bacteria grow, making the food unsafe. Signs like bubbles, bad smells, or changes in color mean the food might be spoiled and should never be tasted. It's safer to throw the jar away to avoid getting sick.

Volunteer notes: Unsealed or improperly processed home-canned foods may harbor dangerous bacteria such as

Clostridium botulinum causing botulism. Visible spoilage and off-odors are clear red flags to discard the food immediately.

Source: SP 50-1000 Turkey basics - "While opening the jar, watch for spurting liquid or cotton-like mold growth on food surfaces or under the lid. Smell for unnatural or off odors. Never taste food from a jar with an unsealed lid or food that shows signs of spoilage."

FS-W03 **IT DEPENDS** *Tweens (10-14)*

You're at a picnic where potato salad has been sitting out at 85°F for about 3 hours. Your grown-up wonders if it's okay to eat later.

Why: Food like potato salad is perishable and should not be left in the 'danger zone' between 40°F and 140°F for more than 2 hours. It depends on how long the salad was out and if it was kept cold enough. If it has been longer than 2 hours at warm temperatures, it's safer to throw it out to avoid getting sick.

Volunteer notes: Perishable foods must not be at temperatures between 40-140°F longer than 2 hours, or 1 hour if above 90°F. Bacteria grow fastest between 60-125°F. Safe handling depends on time and temperature history of the food.

Source: SP 50-1000 Turkey basics - "Perishable food shouldn't be held at lukewarm temperatures (60 to 125 °F) longer than 2 to 3 hours. If perishable foods haven't been kept at the right temperature, throw leftovers out."

FS-T01 **SAFE** *Teens (14-18)*

You washed your hands with soap and water for 20 seconds before preparing a salad with fresh fruits and vegetables, rinsed all produce well under running water, and kept the salad refrigerated below 40°F until serving.

Why: Washing hands thoroughly and rinsing fresh produce reduce bacteria and dirt that can cause illness. Keeping the salad refrigerated below 40°F slows harmful bacterial growth. These steps follow good food safety practices to prevent foodborne illness.

Volunteer notes: Thorough hand washing (20 seconds with soap) and rinsing fresh fruits and vegetables under running water help reduce contamination. Refrigeration below 40°F keeps perishable foods safe by slowing bacterial growth. This follows basic food safety guidelines to prevent illness from contaminated produce and poor hygiene 5:sp_50_837_kitchen_cleanliness__page-001.txt3:sp_50_814_summer_foodsafety__page-001.txt.

Source: SP 50-1000 Turkey basics - "Thoroughly rinse fresh fruits and vegetables under running water; Wash your hands with soap and warm water for 20 seconds before food prep; Keep foods at safe temperatures below 40°F"

FS-T02 **NOT SAFE** *Teens (14-18)*

You left cooked chicken salad out on the picnic table in summer heat (about 85°F) for 4 hours before serving it to your friends.

Why: Cooked, protein-rich foods like chicken salad can grow harmful bacteria very quickly when left out above 40°F for more than 2 hours. At 85°F, bacteria multiply fast, making the salad unsafe to eat and risking foodborne illness.

Volunteer notes: Perishable foods left in the Danger Zone (40°F - 140°F), especially above 60°F, should not be kept out longer than 2 hours. Beyond that, bacterial growth can cause foodborne illness. This violates the 2-hour rule critical for summer food safety .

Source: SP 50-1000 Turkey basics - "Adhere to the 2 hour rule: discard perishables left at room temperature for more than 2 hours; Moist, protein-rich foods need careful handling to avoid rapid bacterial growth"

FS-T03 **IT DEPENDS** *Teens (14-18)*

You canned green beans using a boiling water bath canner because you added lemon juice to make them slightly acidic.

Why: Green beans are low-acid vegetables and typically require pressure canning to destroy harmful bacteria like Clostridium botulinum. Adding lemon juice raises acidity but may not be enough to safely process green beans in a boiling water canner. Safety depends on the exact pH and following tested guidelines for acidity and processing time.

Volunteer notes: Low-acid foods (pH above 4.6), such as most vegetables and meats, need pressure canning for safety because boiling water bath alone may not destroy dangerous bacterial spores. Adding acidifiers like lemon juice can change safety requirements, but only if they lower pH below 4.6 and processing follows tested recipes. Canning safety depends on acidity and processing method to prevent botulism .

Source: SP 50-1000 Turkey basics - "If the food is low in acid (pH 4.6 and above), we are more concerned about safety; Acidic foods include fruits, tomatoes and properly pickled foods; Vegetables are low acid and require pressure canning"

FS-A01 **SAFE** *Adults*

You prepare a fruit jam by following a tested recipe specifically designed for canning, ensuring the pH is below 4.6, processing jars in a boiling water bath for the recommended time, and leaving the proper headspace.

Why: Following a tested recipe for high-acid foods like fruit jam ensures the right acidity and processing time to prevent spoilage and unsafe bacteria growth. Proper headspace and boiling water bath processing are critical for safety. This method destroys molds, yeasts, and most bacteria, making the product safe for home storage.

Volunteer notes: Use only tested recipes with correct pH (<=4.6) and processing times for boiling water bath canning. Proper headspace (usually 1/4 to 1/2 inch) avoids spoilage. Confirm acidity and processing time per OSU/USDA guidance.

Source: SP 50-1000 Turkey basics - "Low acid foods (pH above 4.6) require pressure canning; high acid foods can be safely processed in boiling water - following tested recipes minimizes risks."

FS-A02 **NOT SAFE** *Adults*

You stored raw chicken for 4 hours in a cooler, but the temperature inside rose to 65°F because the ice melted. The chicken smells off and is sticky to the touch.

Why: Raw poultry stored above 40°F for more than 2 hours is unsafe due to rapid bacterial growth in the danger zone (40-140°F). A foul odor and sticky texture also indicate spoilage. Consuming it risks foodborne illness.

Volunteer notes: According to OSU Extension, the 2-hour rule warns against leaving perishable foods above 40°F for over 2 hours. The danger zone promotes growth of pathogens such as Salmonella. Always keep raw poultry cold (below 40°F) during storage and transport.

Source: SP 50-814 Summer food safety - "Discard any perishables left at room temperature for more than 2 hours; moist protein-rich foods like chicken support rapid bacterial growth."

FS-A03 **IT DEPENDS** *Adults*

You plan to freeze home-canned green beans stored in a home freezer at 0°F. The beans might have been processed only in a boiling water bath, and you are unsure whether they were pressure canned.

Why: Freezing home-canned low-acid vegetables is safe only if they were pressure canned initially to destroy Clostridium botulinum spores. If processed only in a boiling water bath, freezing does not eliminate spores and the food may be unsafe. Confirm processing method before freezing.

Volunteer notes: Low-acid vegetables require pressure canning for safety. Freezing affects quality, but safety depends on original processing. Boiling water processing alone is not safe for low-acid foods. Check the original recipe and process before freezing.

Source: SP 50-695 Safety of canned food that freezes - "Low acid foods must be pressure canned; freezing won't destroy botulinum spores if they survive processing."

CANNING FRUITS

CF-K01 **SAFE** *Kids (under 10)*

You watch Grandma put fresh, firm peaches into clean jars. After she finishes, she puts new lids on the jars and carefully makes sure the lids stay flat and don't pop up when she presses them.

Why: This is safe because Grandma used fresh fruit and clean jars, and she made sure the lids were on tight and didn't move. This helps keep the peaches safe to eat.

Volunteer notes: Safe canning of fruits requires using fresh, firm fruit, clean equipment, new lids, and boiling water processing to destroy microorganisms. Checking lids after cooling ensures a good vacuum seal. Improper seals risk spoilage and food waste. Safety relies on following directions exactly (PNW 199 Canning fruits).

Source: PNW 199 Canning fruits - "Standard Mason jars are the best choice for canning. ... To ensure proper sealing, do not use lids that are old, dented, deformed ... Spoilage could result if jars don't seal ... The directions ... have been carefully researched for safe home canning."

CF-K02 **NOT SAFE** *Kids (under 10)*

You notice a jar of canned cherries in the cupboard with a lid that looks bulged and when you press it, the lid pops up. The jam inside looks cloudy and smells funny.

Why: This is not safe because the bulging lid and popping mean air got in or germs grew inside. The cloudy liquid and bad smell are signs the fruit is spoiled or has harmful germs. Eating this fruit could make someone very sick.

Volunteer notes: Signs of spoilage like lids that bulge or pop, cloudy liquid, mold, or off odors mean the canned fruit is unsafe and should be thrown away. Botulism toxin or other bacteria could be present despite normal look or taste, so never taste spoiled jars (PNW 199).

Source: PNW 199 Canning fruits - "With the jar at eye level, look for cloudy canning liquid, rising air bubbles, or any unnatural color. ... Smell for unnatural or off odors. Never taste food from a jar with an unsealed lid or food that shows signs of spoilage."

CF-K03 IT DEPENDS *Kids (under 10)*

You see a jar of figs on the shelf that Grandma made. The lid is closed tight, and the fruit looks normal. But you don't know if Grandma added something special to keep the fruit safe.

Why: Sometimes, the jar of fruit is safe to eat if it has something that helps keep it fresh and the lid is closed well. Other times, if that special thing wasn't added, tiny germs could grow inside, and the fruit might not be safe. When you're not sure if the fruit is safe, it's best to ask an adult before eating.

Volunteer notes: Some fruits like figs require added acid for safe boiling water canning because their natural acid is too low to prevent harmful bacteria growth. If acidification and proper processing were done, the sealed jar is safe. Without added acid, canned figs may be unsafe (PNW 199).

Source: PNW 199 Canning fruits - "Some fruits, such as Asian pears and figs, require the addition of acid (i.e., lemon juice) to be safely processed in a boiling water canner."

CF-W01 SAFE *Tweens (10-14)*

You helped a grown-up pick fresh, firm peaches and wash them. You watched while the grown-up carefully packed the peaches into jars and handled the hot water part safely. After they finished, they sealed the jars following the recipe instructions.

Why: When a grown-up follows the recipe and seals the jars properly, canning peaches is safe. This is because peaches have natural acids that help keep the food safe inside the jar.

Volunteer notes: Fruits are high-acid foods and can be safely processed in a boiling water canner. Proper cleaning, using new lids, correct processing times, and sealed jars are essential for safety and quality. This prevents dangerous bacteria and spoilage microorganisms.

Source: PNW 199 Canning fruits - "When canning fruits, you can safely use a boiling water canner. Molds, yeasts, and bacteria which can grow in these high-acid foods are destroyed at boiling water-bath temperatures."

CF-W02 NOT SAFE *Tweens (10-14)*

You see a grown-up helping to put apples into jars at home. They use some old lids that look a bit bent and don't close the jars very tightly. After putting the jars away, a few of them didn't close all the way, but the jars were left out on the kitchen counter for a long time.

Why: When jars have old or bent lids and aren't closed tightly, air and germs can get inside. This can make the apples go bad and unsafe to eat. It's important for grown-ups to use new lids and make sure the jars are sealed well before putting them away.

Volunteer notes: Lids that are old, dented, or reused are not reliable for sealing jars. Jars must be sealed to prevent spoilage and foodborne illness. Storing unsealed canned fruits at room temperature is unsafe. If jars don't seal, they must be refrigerated and eaten soon or reprocessed with new lids.

Source: PNW 199 Canning fruits - "Do not use lids that are old, dented, deformed, have gaps or defects. Spoilage could result if jars don't seal, and food is wasted."

CF-W03 IT DEPENDS *Tweens (10-14)*

You found a jar of homemade canned pears in the pantry with a lid that looks sealed, but it was not processed long enough according to the recipe. The jar was stored in a cool, dark place.

Why: Even if the jar looks sealed and was stored well, not processing the pears long enough can mean harmful bacteria or toxins might still be in the jar. The safety depends on whether the processing time was enough to kill all bacteria. If not, it's best not to eat the fruit and instead reprocess or throw it away.

Volunteer notes: Proper processing time is critical to destroy microorganisms in canned fruits. A sealed lid alone does not guarantee safety if the processing time was too short. Incomplete processing can allow bacteria or toxins to survive, causing food poisoning risk. Storage conditions matter but do not fix poor processing steps.

Source: PNW 199 Canning fruits - "Foods that were under-processed or improperly processed and held over 24 hours should be destroyed."

CF-T01 **SAFE** *Teens (14-18)*

Jamie is canning fresh peaches using a boiling water canner. She selects ripe, firm peaches, washes them thoroughly, and packs them into hot, sterilized jars. Jamie follows a tested recipe, adding the recommended amount of lemon juice for acidity and processes the jars for the exact time specified to ensure safety.

Why: Because peaches are acidic fruits, they can be safely canned using a boiling water canner if you follow a tested recipe exactly. Adding lemon juice maintains the necessary acidity to prevent harmful bacteria growth. Processing the jars for the right time at boiling temperatures destroys yeasts, molds, and bacteria that can spoil the fruit or cause illness.

Volunteer notes: Fruits are high-acid foods and safe for boiling water bath canning if processed according to research-tested recipes, including acid additions like lemon juice when needed. Processing times and temperatures destroy spoilage organisms and pathogens to ensure safety. A boiling water canner reaching 212°F is sufficient for high-acid fruits like peaches .

Source: PNW 199 Canning fruits - "When canning fruits, you can safely use a boiling water canner. Some fruits, such as Asian pears and figs, require the addition of acid (i.e., lemon juice) to be safely processed in a boiling water canner."

CF-T02 **NOT SAFE** *Teens (14-18)*

Alex decides to can fresh strawberries but uses old lids he found in the back of the cupboard. He did not wash the jars or lids, skipped preheating the jars, and processed the jars for less time than recommended in the recipe.

Why: Using old or damaged lids can prevent jars from sealing properly, allowing air and bacteria inside. Not washing jars and lids risks contamination. Processing for less time than recommended may leave harmful microorganisms alive. These mistakes can cause spoilage or serious foodborne illness, so the canned fruit is unsafe to eat.

Volunteer notes: Proper canning requires using new or good-condition lids for sealing, thoroughly washed and preheated jars, and following exact processing times to destroy bacteria. Inadequate heat or poor sealing leads to spoilage and possible food safety hazards, including botulism in some cases. Even if the food looks and smells fine, it may be unsafe .

Source: PNW 199 Canning fruits - "Inspect jars for cracks and chips, and discard damaged ones. To ensure proper sealing, do not use lids that are old, dented, deformed, or have been used before. Following the directions exactly is vital to ensure a product that is top quality and safe to eat."

CF-T03 **IT DEPENDS** *Teens (14-18)*

Sara wants to can blueberries using a boiling water canner but didn't add any lemon juice or other acid. She wonders if her canned blueberries will be safe since they are naturally sweet and have some acidity.

Why: Blueberries are acidic fruits, but the acid level can vary. Generally, boiling water canning is safe for blueberries without added acid, but some recipes recommend adding lemon juice to ensure proper acidity especially if the blueberries are less ripe or grown in low-acid conditions. Safety depends on following a tested recipe that addresses acid levels and processing times.

Volunteer notes: Some fruits like blueberries are naturally acidic enough to be safely canned in boiling water canners without added acid, but acid additions may be recommended based on ripeness and variety to guarantee safety. Always use research-tested recipes that specify acid additions and processing times to prevent microbial growth and spoilage. When in doubt, add acid or use a tested recipe for safety .

Source: PNW 199 Canning fruits - "Some fruits, such as Asian pears and figs, require the addition of acid (i.e., lemon juice) to be safely processed in a boiling water canner."

CF-A01 **SAFE** *Adults*

You prepare quart jars of fresh peaches using a tested boiling water canner recipe, leaving 1/2 inch headspace, use new lids, process at a rolling boil for the recommended time, and cool jars at room temperature without retightening lids.

Why: Following a tested recipe with appropriate headspace, using a boiling water canner for high-acid fruit like peaches, and proper cooling without retightening lids ensures safe processing. Using new lids and processing for full recommended time prevents spoilage and unsafe conditions.

Volunteer notes: Boiling water canning is safe for high-acid fruits like peaches when you follow tested recipes exactly, ensuring correct headspace and process time. Retightening lids after processing can cause seal failure by damaging the gasket. Always cool jars at room temperature without cold drafts. This practice prevents spoilage and ensures vacuum seals are secure.

Source: PNW 199 Canning fruits - "Follow the preparation instructions carefully. Begin counting the processing time when the water comes back to a rolling boil. Various altitudes. Adjust the processing time for elevations over 1,000 feet. Headspace. Leave 1/2 inch headspace for *both the fruit and liquid unless otherwise specified. ... Do not retighten screw bands after processing. Retightening of hot lids may cut through the gasket and cause seal failure."*

CF-A02 **NOT SAFE** *Adults*

You canned strawberries in wide-mouth jars but washed and reused old lids multiple times. After processing, many jars did not seal properly, and some jars showed cloudy juice and bubbles after storage.

Why: Reusing old lids can prevent proper sealing because the sealing compound degrades after one use. Cloudy juice and bubbles inside jars are signs of spoilage. Improper seals increase risk of microbial growth and unsafe food. Such jars should be refrigerated and used quickly or discarded.

Volunteer notes: Use new lids every time to ensure proper seal. Old or reused lids may not seal correctly, risking spoilage and foodborne illness. Cloudy liquid and bubbles indicate microbial activity. If jars fail to seal, reprocess or refrigerate and consume soon. Spoiled home canned food should be discarded to avoid foodborne illness.

Source: PNW 199 Canning fruits - "To ensure proper sealing, do not use lids that are old, dented, deformed, have gaps or other defects in the sealing gasket, or have been used before. Spoilage could result if jars don't seal, and food is wasted. ... look for cloudy canning liquid, rising air bubbles, or any unnatural color."

CF-A03 **IT DEPENDS** *Adults*

A home canner wants to preserve Asian pears by boiling water canning but is unsure about adding acid and processing times. They plan to use a water bath canner but do not know the altitude adjustments.

Why: Asian pears have borderline acidity and require adding acid (such as lemon juice) to be safe when boiling water canning. Processing times and altitude adjustments must also be accurate. Without these specifics, safety cannot be assured. Proper acidification and time adjustments are critical to prevent microbial growth.

Volunteer notes: Some fruits, like Asian pears, need added acid for safe boiling water canning because of lower natural acidity. Processing times depend on altitude; higher elevations require longer times. Not adding acid or adjusting time correctly risks spoilage and botulism. Always follow tested recipes for such borderline foods.

Source: PNW 199 Canning fruits - "Some fruits, such as Asian pears and figs, require the addition of acid (i.e., lemon juice) to be safely processed in a boiling water canner. ... Adjust the processing time for elevations over 1,000 feet."

CANNING TOMATOES & SALSAS

CT-K01 **SAFE** *Kids (under 10)*

You see a jar of salsa on the shelf with a lid that is tightly sealed and flat on top, and it smells fresh and a little tangy when you sniff it carefully.

Why: If the lid is tightly sealed down and the jar smells fresh without any strange or bad smells, the salsa is safe to eat. The seal means air and germs couldn't get in, so the food stays good. Always check that jars are sealed well before opening.

Volunteer notes: A jar with a flat, unpopped lid that doesn't move when pressed and has no off-odors or mold is considered safely canned. These signs confirm a proper vacuum seal preventing botulism and spoilage in acidified salsas processed in boiling water canners .

Source: PNW 172 Canning vegetables - "Test each jar for a vacuum seal. Jars with flat, metal lids are sealed if the lid has popped down in the center or does not move when pressed down. Smell for unnatural or off odors."

CT-K02 **NOT SAFE** *Kids (under 10)*

You see a jar of tomatoes in the kitchen, but the lid is sticking up like it's puffed out. When the jar is opened, it smells yucky and rotten.

Why: When a jar's lid looks all puffy and the food inside smells bad, it means germs might have grown inside. This food is not safe to eat because it can make you sick.

Volunteer notes: Bulging lids and off-odors indicate possible botulism toxin or spoilage. These signs mean the food was not processed correctly or the jar lost its seal, making the canned tomatoes unsafe to eat .

Source: PNW 172 Canning vegetables - "When commercially canned food shows any sign of spoilage-bulging can ends, off-odor, or mold-do not use it."

CT-K03 **IT DEPENDS** *Kids (under 10)*

You see a jar of homemade salsa on the counter that was left out overnight, but when you look at it, there is no mold or funny smell on it.

Why: Leaving salsa out all night is risky because harmful germs can grow even if it looks and smells okay. If it was canned correctly and sealed well and is a tested recipe, then it might still be okay. But if it was not processed properly or the jar seal is broken, it's not safe.
Volunteer notes: Safety depends on proper canning method, acidity, and whether the jar seal was intact. Salsa must be made from tested recipes with correct acid levels and processed in boiling water canners. Unsealed or improperly processed jars or those exposed to warm temperatures risk botulism .
Source: PNW 172 Canning vegetables - "Salsas must be acidified enough to prevent botulism. Leaving jars out can risk safety unless sealed and canned exactly as tested."

CT-W01 SAFE *Tweens (10-14)*

You watched a grown-up follow a trusted salsa recipe carefully. They added bottled lemon juice and the right amount of vinegar, mixed everything well, then heated the jars using a special pot to make sure they were safe and sealed tight.

Why: This salsa is safe because the grown-up used a recipe that's been tested to keep food safe. The lemon juice and vinegar help stop bad germs from growing, and heating the jars properly makes sure they're sealed well so the salsa stays fresh.
Volunteer notes: Safe salsa canning requires following research-tested recipes that ensure adequate acidity and processing using a boiling water canner to prevent botulism. Acid sources like bottled lemon juice or 5% vinegar are critical. Proper sealing and processing times must be followed exactly to ensure safety .
Source: PNW 395 Salsa recipes for canning - "Salsa recipes for canning must meet acidity requirements to prevent growth of botulism bacteria. Follow directions carefully for safety."

CT-W02 NOT SAFE *Tweens (10-14)*

Your friend helped their family make fresh salsa using an old recipe. But they didn't add any vinegar or lemon juice, which helps keep the salsa safe. They also didn't have a grown-up boil the jars to make sure everything was clean and sealed tightly.

Why: Without vinegar or lemon juice, the salsa might not be safe because some germs can still grow. And if the jars aren't boiled, those germs might not be killed. Even if the salsa looks and smells okay, it can still make you very sick.
Volunteer notes: Salsas must be acidified with vinegar or lemon juice and processed in a boiling water canner to eliminate the risk of Clostridium botulinum. Recipes without acid or proper processing time are unsafe. Skipping steps or using untested recipes poses a health risk .
Source: PNW 395 Salsa recipes for canning - "Never can salsas that do not follow research-tested recipes; low acidity can allow botulism toxin to form."

CT-W03 IT DEPENDS *Tweens (10-14)*

You found a jar of home-canned tomatoes in the fridge without a label. The jar is sealed tightly and the tomatoes look fine, but you don't know who canned it or how.

Why: These tomatoes might be safe if they were canned using a tested recipe with enough acid and processed in a boiling water canner. But if the process wasn't done right, or the acidity is low, harmful bacteria could grow. Without knowing the canning method and if the jar was sealed and stored properly, it's best to be cautious.
Volunteer notes: Safety depends on processing method, acidity, and storage. Tomatoes are acid enough for boiling water canning if acidified properly. Unknown canning method or missing recipe information means you can't be sure of safety. When in doubt, discard or reprocess using research-tested methods .
Source: PNW 395 Salsa recipes for canning - "Safety depends on acidity level and processing method; improper canning or unknown methods can be unsafe."

CT-T01 SAFE *Teens (14-18)*

You canned a batch of salsa following a tested recipe exactly, using bottled 5% white vinegar for acidity and processed the jars in a boiling water canner for the recommended time.

Why: Following a tested recipe ensures your salsa is acidic enough to prevent the growth of harmful bacteria like botulism. Using 5% vinegar and processing properly in a boiling water canner kills bacteria and seals the jars safely. These exact steps are critical for safe salsa canning.
Volunteer notes: Safe salsa canning requires following research-tested recipes that specify acid levels and boil water canner processing times. 5% acidity vinegar and exact processing prevent botulinum toxin risk. Do not alter acid amounts or processing.
Source: PNW 395 Salsa recipes for canning - "Most salsa recipes are a mixture of low-acid foods ... with acid foods, such as tomatoes. Salsa

recipes ... must meet acidity-level requirements to prevent growth of botulism bacteria. ... follow the directions carefully for each recipe ... Never can salsas that do not follow these or other research-tested recipes."

CT-T02 **NOT SAFE** *Teens (14-18)*

You canned fresh tomatoes without adding any lemon juice or citric acid, and processed the jars in a boiling water canner only for 20 minutes.

Why: Tomatoes need added acid-like lemon juice or citric acid-when canned in a boiling water bath because their natural acidity isn't always enough to kill botulism spores. Skipping this step or underprocessing can allow dangerous bacteria to survive.

Volunteer notes: Boiling water canning tomatoes requires adding recommended acid amounts. Without added acid, processing times and safety cannot be assured, risking botulism poisoning from Clostridium botulinum spores surviving.

Source: PNW 300 Canning tomatoes and tomato products - "The addition of acid in the form of lemon juice or citric acid is recommended in all tomato products canned in a boiling water bath as a precautionary measure."

CT-T03 **IT DEPENDS** *Teens (14-18)*

You want to add fresh garlic and extra thickening with tomato paste before canning your salsa in a boiling water canner.

Why: Adding fresh garlic before canning lowers the salsa's acidity, which can let botulism bacteria grow unless the recipe and processing method are specifically tested. Also, thickening before canning can cause unsafe heat penetration. You should only add such ingredients if the recipe is lab-tested to be safe.

Volunteer notes: Fresh garlic or thickeners added before canning salsa affect acidity and heating, risking botulism. Only use tested recipes allowing such ingredients. Otherwise, add these after opening, or freeze instead.

Source: PNW 395 Salsa recipes for canning - "It is not safe to add or increase fresh herbs or garlic before canning because they affect acidity. Do not thicken salsas before canning. Thickening can be done after opening."

CT-A01 **SAFE** *Adults*

You prepare salsa using a current, tested recipe that specifies adding bottled lemon juice and following processing times exactly in a boiling water canner. You use fresh, ripe tomatoes and run the filled jars through the recommended processing time at the correct altitude.

Why: Following a tested recipe that includes the correct amount of acid (like bottled lemon juice or vinegar), and processing the salsa as instructed ensures the acidity is sufficient to prevent botulism. Using fresh, ripe tomatoes and following processing guidelines precisely is critical to safety.

Volunteer notes: Salsa must meet acidity-level requirements and be processed exactly as tested to be safe for boiling water canning. Acidified foods need measured acid to prevent botulism risk; untested recipes are unsafe for canning and should be frozen or refrigerated instead.

Source: PNW 395 Salsa recipes for canning - "Most salsa recipes are a mixture of low-acid foods...with acid foods, such as tomatoes or fruit...Follow the directions carefully for each recipe...Never can salsas that do not follow these or other research-tested recipes."

CT-A02 **NOT SAFE** *Adults*

You canned fresh tomatoes but did not add any acid like bottled lemon juice or citric acid before processing in a boiling water canner. The tomatoes were overripe and came from a frost-damaged vine.

Why: Tomatoes that are overripe or from frost-damaged vines lose acidity, which is necessary for safe boiling water canning. Without adding acid, botulism spores might survive. Not acidifying and using low-quality tomatoes can result in an unsafe product.

Volunteer notes: Tomatoes must be acidified before boiling water canning to lower pH and prevent botulism. Overripe or frost-killed tomatoes lose acidity and should not be canned without adding acid. When canning tomatoes, always adjust acidity and use quality fruit.

Source: PNW 395 Salsa recipes for canning - "Avoid tomatoes that are overripe or from dead or frost-killed vines...These will result in a poor-quality and potentially unsafe product...Use the amount and type of acid the recipe calls for...Breaking this law could be life threatening."

CT-A03 **IT DEPENDS** *Adults*

You have an original salsa recipe you created with fresh tomatoes, peppers, and onions, but you are unsure if the acidity level is sufficient. You want to can it in a boiling water bath to store long-term.

Why: Safety depends on whether the salsa's acidity (pH) is low enough to prevent botulism growth. Without a tested recipe or lab analysis, the acid level is unknown; this means boiling water canning might not be safe unless bottled lemon juice or vinegar is added in a tested amount. Otherwise, freezing or refrigeration is safer.

Volunteer notes: Only follow research-tested salsa recipes for boiling water canning because they guarantee adequate acidity. Untested or original recipes may be unsafe due to unknown pH. When in doubt, freeze or refrigerate instead of canning.

Source: PNW 395 Salsa recipes for canning - "Because salsas are a mixture of acid and low-acid ingredients...salsa recipes must meet acidity-level requirements...Never can salsas that do not follow these or other research-tested recipes. Freezing is the only safe option for preserving untested or original salsa recipes."

LOW ACID FOODS

LA-K01 **SAFE** *Kids (under 10)*

You see jars of canned green beans in the pantry. The jars were packed by an adult using a special cooker that locks tight and gets really hot. The jars have tight lids and no bubbles or strange smells.

Why: Low acid foods like green beans need to be cooked in a special cooker called a pressure canner, which gets hotter than boiling water to kill all bad germs. If the jars are sealed tight and look normal with no strange smells or bubbles, they are safe to eat.

Volunteer notes: Low acid foods like vegetables must be pressure canned to reach 240°F, which kills Clostridium botulinum bacteria. Proper sealing and processing time are critical. This prevents deadly botulism even if the food looks and smells normal. Adults must follow researched processing times exactly.

Source: PNW 361 Canning meat poultry game - "Poultry, red meats, and meat from game animals (such as deer, elk, and bear) are low-acid foods and must be processed in a pressure canner to ensure safety. ... Only pressure canning produces temperatures high enough (240°F, 28 degrees above boiling) to kill many bacteria which can grow in low-acid foods, including Clostridium botulinum."

LA-K02 **NOT SAFE** *Kids (under 10)*

You notice some cooked potatoes left on the kitchen table all day. They feel warm and have started to smell funny when you sniff near them.

Why: Cooked low acid foods like potatoes left out at room temperature too long let bad bacteria grow that can make people very sick. If they smell bad, it's a sure sign bacteria have started making poisons. These foods should be thrown away.

Volunteer notes: Cooked low acid foods held in the danger zone (40-140°F) for more than 2-3 hours can support the growth of Clostridium botulinum and other bacteria. Such foods should be discarded regardless of appearance or smell if left out too long.

Source: PNW 450 Canning smoked fish at home - "Watch for these potential food safety concerns - Low acid foods (especially cooked ones) that have been kept in the danger zone (40 - 140 °F - and especially 60 - 125 °F) for more than 2 - 3 hours."

LA-K03 **IT DEPENDS** *Kids (under 10)*

You see a jar of homemade canned vegetables on the shelf. The lid looks a little popped up. The jar had been left unopened for a few months, but you also notice a little fuzzy mold growing around the lid edges.

Why: A popped lid or mold on canned low acid foods usually means the food might not be safe. But if the mold is only on the outside and no bad smells or bubbles are inside, an adult might be able to carefully clean it off and check. If the food smells funny, is discolored, or if the lid is bulging, it must be thrown away to stay safe.

Volunteer notes: A bulging lid or mold often signals unsafe conditions in low-acid canned foods. External mold might not always mean spoilage, but the presence of bad odors, gas bubbles, or lid bulging indicates microbial growth. The safest advice is to discard questionable jars due to botulism risk.

Source: PNW 450 Canning smoked fish at home - "What does the food look like and smell like? Are there any visible signs that the food might be spoiled (such as mold, bubbles rising, and discoloration)?"

LA-W01 **SAFE** *Tweens (10-14)*

You watch your grown-up cook ground beef and put it into special jars using a special kitchen tool that makes sure the jars are safe. They follow the recipe carefully and use the right steps. After sealing the jars, the grown-up stores them in a cool, dark place.

Why: Meat needs to be prepared with the right tools and steps to keep it safe. Since your grown-up follows the recipe carefully and stores the jars properly, the food will be safe to eat.

Volunteer notes: Low acid foods (pH above 4.6) such as meat, poultry, and fish must be processed using a pressure canner to safely kill Clostridium botulinum spores. Boiling water canners do not reach high enough temperatures. The recipe and process must be followed exactly for safety. Afterward, store sealed jars in a cool place. Common question: Why can't I just use a boiling water canner for meat? Because it doesn't get hot enough to kill dangerous bacteria in low acid foods.

Source: PNW 361 Canning meat poultry game - "Poultry, red meats, and meat from game animals are low-acid foods and must be processed in a pressure canner to ensure safety. Only pressure canning produces temperatures high enough to kill many bacteria which can grow in low-acid foods, including Clostridium botulinum."

LA-W02 NOT SAFE *Tweens (10-14)*

You see some jars of meat stew in the kitchen that a grown-up put away a long time ago. The jars look normal, but you don't know if the grown-up used the right special method to keep the stew safe. Meat stew needs to be preserved using a special tool that gets hotter than just boiling water. If the grown-up didn't use this special method, the stew might not be safe to eat, even if the jars look fine.

Why: Meat stew is a type of food that needs a special way to be preserved safely. This special method uses a tool that gets hotter than boiling water to make sure all harmful germs are gone. Just using boiling water isn't enough to keep meat stew safe to eat. So, if a grown-up didn't use the right special tool, the stew might still be unsafe, even if the jars look okay.

Volunteer notes: Boiling water or steam canners don't reach the necessary temperature (240°F) to kill Clostridium botulinum spores in low acid foods. Only pressure canners do. Eating improperly canned low acid foods risks deadly botulism poisoning. Likely question: What if the food looks and smells okay? Botulism toxin doesn't change how food looks or smells, so you can't tell by sight or smell if it's dangerous.

Source: PNW 361 Canning meat poultry game - "Boiling water canners and steam canners do not produce temperatures high enough to kill botulism-causing bacteria and other spoilage organisms. Never can meat products or combination foods for which you do not have research-based processing times."

LA-W03 IT DEPENDS *Tweens (10-14)*

You notice your family canned green beans using a different method than usual. They didn't use the special cooker meant for low acid foods, but after canning, they put the jars straight into the fridge. You wonder if the green beans are still safe to eat.

Why: Green beans need careful cooking to be stored safely for a long time because they don't have much acid. If your family didn't use the special cooker but cooled the jars quickly in the fridge, the beans might be safe only for a few days. After that, they might not be safe to eat, so it's important to check with a grown-up before trying them.

Volunteer notes: Low acid foods canned without a pressure canner can support dangerous bacteria like botulism unless kept very cold. Refrigeration slows bacteria but does not make the food shelf-stable. Such foods must be stored in the fridge and eaten quickly. Not processing the food properly means it's not shelf stable and could be unsafe if stored at room temperature. Common question: Can refrigeration fully keep low acid food safe if not pressure canned? It helps short-term, but doesn't replace safe canning.

Source: PNW 361 Canning meat poultry game - "Due to the risk of botulism, never consume low-acid foods that are not canned according to the recommendations... Foods kept cold may reduce risk if consumed soon, but must not be stored at room temperature."

LA-T01 SAFE *Teens (14-18)*

You just finished pressure canning green beans following the exact USDA-approved recipe with the right pressure and time. The jars sealed properly and you cooled the canner naturally before removing the food.

Why: Low acid foods like green beans can only be safely canned using a pressure canner because it reaches 240°F, enough to kill dangerous bacteria like Clostridium botulinum. Following tested recipes and processing times ensures safety. If the jars sealed properly and the canner cooled down naturally, the food is safe to store and eat.

Volunteer notes: Low acid foods must be processed in a pressure canner at 240°F for a precise time to kill botulism spores. Natural cool down is part of the heat kill process. Home canning must follow research-tested times and pressure (PNW 361).

Source: PNW 361 Canning meat poultry game - "Because meat and poultry products are low-acid foods, Clostridium botulinum can survive and grow unless pressure canned at 240°F. Follow USDA recipes exactly."

LA-T02 NOT SAFE *Teens (14-18)*

Someone packed quart jars very full with cooked ground beef, skipped the 10 minute venting step on the pressure canner, and then canned at home using a boiling water bath instead of a pressure canner.

Why: Boiling water bath canners don't reach the high temperature needed to destroy botulism spores in low acid foods

like meat. Over-packed jars heat unevenly, and skipping the venting step traps cold air which prevents proper heating. This risks deadly botulism.
Volunteer notes: Never use boiling water or steam canners for low acid foods. Pressure canners reaching 240°F and proper venting are required to kill dangerous bacteria. Overpacked jars don't heat evenly, increasing risk (PNW 361).
Source: PNW 361 Canning meat poultry game - "Boiling water canners do not produce temperatures high enough to kill botulism-causing bacteria. Always exhaust air by venting for 10 minutes before processing."

LA-T03 IT DEPENDS *Teens (14-18)*

You want to store cooked low acid vegetables like green beans in quart jars but are unsure if you should pressure can or freeze them first.

Why: Low acid cooked vegetables need pressure canning to be safe for shelf storage because the heat kills botulism spores. However, if you freeze them correctly instead, freezing keeps bacteria from growing, making the food safe without canning. So safety depends on the preservation method chosen.
Volunteer notes: Low acid foods require pressure canning for shelf stability. Freezing is a safe alternative preservation but requires keeping food frozen continuously to prevent bacterial growth (SP 50 493).
Source: PNW 450 Canning smoked fish at home - "Low acid foods like vegetables, meats must be pressure canned to kill bacteria for shelf safety. Alternatively, freezing controls microbial growth by low temperature."

LA-A01 SAFE *Adults*

You pressure-can diced potatoes at sea level following a research-tested recipe exactly, adjusting pressure and time as directed. You use a dial-gauge pressure canner that has been recently tested for accuracy.

Why: Low-acid foods like potatoes must be processed in a properly operated pressure canner to reach 240°F and kill Clostridium botulinum spores. Using a tested recipe with correct pressure, time, and altitude adjustments ensures safety. An accurate gauge ensures the correct temperature is reached and maintained during the entire process.
Volunteer notes: Low-acid foods require pressure canning at 240°F. Follow research-based recipes exactly. Adjust pressure for altitude and ensure your gauge is accurate to prevent botulism risk. This aligns with USDA/OSU Extension guidelines.
Source: PNW 361 Canning meat poultry game - "Due to the risk of botulism, always can meats and other low-acid foods in a pressure canner; follow research-based recipes and adjust pressure for altitude. Check gauge accuracy yearly."

LA-A02 NOT SAFE *Adults*

You prepare homemade garlic-in-oil at room temperature without acidifying the garlic or refrigerating the oil mixture. You plan to store it on the counter for several weeks to use as a topping.

Why: Garlic in oil is a low-acid, anaerobic environment ideal for Clostridium botulinum growth if not acidified or refrigerated. Storing it at room temperature without acidification can lead to deadly botulism toxin formation, even if it looks and smells normal.
Volunteer notes: Low-acid foods in oil must be acidified and/or refrigerated to prevent botulism. Commercial versions are acidified for safety. Homemade versions need acidification per tested procedures or refrigeration for only 4 days to reduce risk.
Source: PNW 450 Canning smoked fish at home - "Low acid foods stored in oil create conditions for Clostridium botulinum; acidification or refrigeration is essential for safety."

LA-A03 IT DEPENDS *Adults*

You canned a vegetable soup with meat in quart jars using a boiling water canner because the recipe did not specify pressure canning. You processed the jars for 90 minutes at boiling temperature.

Why: Vegetable soups with meat are low-acid foods that require pressure canning to reach 240°F to kill botulism spores. Boiling water canning (212°F) is insufficient. Safety depends on whether the recipe is research-tested for boiling water canning (rare for such soups). Without tested instructions, pressure canning is mandatory. Altitude and jar size also affect processing time.
Volunteer notes: Low-acid mixed foods must be pressure canned following tested recipes. Boiling water canners do not reach safe temperatures for these foods. When in doubt, do not use boiling water canning for meat-based soups.
Source: PNW 421 Use care operation of your pressure canner - "Only pressure canners produce the temperature (240°F) needed to safely process low-acid foods including meat-vegetable mixtures; boiling water canners cannot."

PICKLING

PK-K01 **SAFE** *Kids (under 10)*

You found a jar of pickles in the kitchen. The lid is tight and popped down in the middle. The jar looks clear inside and smells fresh like vinegar.

Why: When a jar lid is sealed tight and pressed down in the center, it shows the pickles are safely preserved. Clear liquid and a fresh vinegar smell mean the pickles are good to eat. Always check for a strong sealed lid and no mold or bad smells before eating pickles from a jar.

Volunteer notes: A proper vacuum seal is indicated by a lid that is concave and does not move when pressed. Clear brine and vinegar smell confirm safe acid levels preventing bacterial growth. Use tested recipes with 5% vinegar acidity for safety. A sealed jar with no visible spoilage is safe to eat.

Source: SP 50-466 Pickle fact sheet - "Test each jar for a vacuum seal. Jars with flat metal lids are sealed if the lid has popped down in the center and does not move when pressed."

PK-K02 **NOT SAFE** *Kids (under 10)*

You see a jar of cucumbers that someone left on the kitchen counter for three days. The jar has no lid and the cucumbers look slimy and smell bad.

Why: Leaving pickles without a lid on the counter for days lets germs grow. Slimy cucumbers and bad smells mean the pickles went bad and could make you sick. Always keep pickles in the fridge with a lid to stay safe.

Volunteer notes: Pickles left unrefrigerated and uncovered for long periods enter the temperature danger zone, allowing harmful bacteria to grow. Sliminess and foul odors are signs of spoilage. Foods not stored properly should be discarded to avoid foodborne illness.

Source: SP 50-834 Tips for critical thinking - "Temperature is very important. If the product has been in the danger zone (40 - 140 °F) for longer than 2 - 3 hours, it could be a safety concern."

PK-K03 **IT DEPENDS** *Kids (under 10)*

You see a jar of pickles in the fridge with a tiny bit of white fuzzy stuff on top. The jar hasn't been opened yet. Can the pickles still be okay to eat?

Why: Sometimes tiny white fuzzy spots can grow even in the fridge, but that might mean the pickles are not safe anymore. If you see fuzzy stuff, smell something funny, or the jar lid isn't closed tightly, it's best to tell an adult and not eat the pickles. But if the jar is closed tightly and everything else looks and smells okay, sometimes grown-ups carefully take off the fuzzy part and keep eating the rest.

Volunteer notes: Mold on pickles can signal spoilage especially if accompanied by off-odors or broken seals. In fermented or acid-pickled foods, surface mold is uncommon but possible if storage or sanitation was poor. USDA advises discarding moldy pickles to avoid risk of toxins or bacteria. Safety depends on mold presence and product condition.

Source: UC Davis Common issues with fermented fruits and vegetables - "Watch for signs of spoilage like mold, bubbles rising, and discoloration; if mold is present, the safest choice is to discard the product."

PK-W01 **SAFE** *Tweens (10-14)*

You helped your family make quick pickles by soaking cucumbers in vinegar with at least 5% acidity, following the recipe exactly, and storing the jar in the fridge.

Why: Quick pickles are safe when you use vinegar with at least 5% acidity and follow the recipe exactly, especially the vinegar amount. Keeping pickles refrigerated also stops harmful bacteria from growing.

Volunteer notes: Quick pickles must have vinegar with 5% acidity or higher and proper storage in the refrigerator ensures safety. Always follow tested recipes and do not reduce vinegar amounts.

Source: PNW 355 Pickling vegetables - "The key ingredient that ensures safety in quick pickles is the vinegar. Use any vinegar with 5% acidity. Do not reduce the amount of vinegar in a recipe. Keep under refrigeration for safety."

PK-W02 **NOT SAFE** *Tweens (10-14)*

Someone left a jar of homemade pickled fish on the kitchen counter for a week without refrigerating it or using enough vinegar in the pickling solution.

Why: Pickled fish must be stored in the refrigerator and picked with enough vinegar. Leaving it unrefrigerated for a week or with too little vinegar lets dangerous bacteria grow, making it unsafe to eat.

Volunteer notes: Pickling fish requires refrigeration below 50F and a vinegar-to-water ratio of at least 1:1 to prevent harmful bacteria. Not following these steps risks botulism and food poisoning.

Source: PNW 183 Pickling fish and other aquatic foods - "Fish must be stored under refrigeration (38F) and pickling solutions should not have less than equal parts vinegar to water to be safe."

PK-W03 **IT DEPENDS** *Tweens (10-14)*

Your older sibling made fermented pickles by soaking cucumbers in salted water, but you're not sure if they rinsed off the pickling lime well enough before putting the cucumbers into the jar.

Why: Using pickling lime to firm cucumbers can be safe only if the cucumbers are rinsed 3 times to remove extra lime. If not rinsed well, the lime can cause health problems, so you need to check if rinsing was done correctly.

Volunteer notes: Pickling lime must be completely rinsed off cucumbers through multiple soaks and rinses to remove excess calcium lime, which is toxic if ingested. If rinsing is incomplete, the pickles are unsafe.

Source: PNW 355 Pickling vegetables - "Cucumber slices soaked in lime solution must be rinsed repeatedly to remove excess lime. Do not use lime from garden centers. Excess lime must be removed for safety."

PK-T01 **SAFE** *Teens (14-18)*

You followed a tested quick-pickle recipe that uses vinegar with 5% acidity and fresh vegetables. You kept the jars in the refrigerator and ate them within a few weeks.

Why: This is safe because the vinegar acidity level is high enough to prevent harmful bacteria. Using a tested recipe ensures the right balance of ingredients, and refrigeration slows spoilage. Eating within a few weeks reduces risk of spoilage.

Volunteer notes: USDA and OSU guidelines require vinegar of at least 5% acidity for safety in quick pickles. Refrigeration and short storage times further protect against spoilage and pathogens.

Source: PNW 355 Pickling vegetables - "Use any vinegar with 5% acidity...Do not reduce the amount of vinegar or increase the amount of water in a recipe...Follow instructions for conventional processing or use lower-temperature pasteurization."

PK-T02 **NOT SAFE** *Teens (14-18)*

A friend decided to make pickles using homemade vinegar with unknown acidity, reducing the vinegar amount and doubling the water, then canned the pickles without processing.

Why: Using homemade vinegar of unknown acidity and diluting the vinegar makes the pickle low acid. Without enough acid or proper heat processing, harmful bacteria like Clostridium botulinum can grow, causing serious illness.

Volunteer notes: Pickles must have sufficient acidity (pH below 4.6) typically ensured by 5% vinegar; homemade or diluted vinegar is unsafe. Proper heat processing is required to kill pathogens. This situation risks botulism.

Source: NCHFP Pickled eggs - "Do not use homemade vinegar...The percent acidity must be known for successful pickling...Vinegar is important for safety reasons."

PK-T03 **IT DEPENDS** *Teens (14-18)*

You make fermented pickles by packing cucumbers into a jar with salt brine and leaving them at room temperature to ferment. You wonder if it is safe to eat after 1 day.

Why: Safety depends on whether enough acid has developed from fermentation to lower pH below 4.6, which usually takes several days. Early consumption can be unsafe because harmful bacteria might still grow if acidity is insufficient.

Volunteer notes: Fermented pickles are safe only once acid is produced to inhibit pathogens; this usually requires several days at the right temperature. Following tested recipes and waiting for proper fermentation is critical.

Source: UC David Safely fermented fruit and vegetable - "Botulism can occur if fermentation is not successful and acid levels are low...The production of sufficient acid will inhibit the growth of Clostridium botulinum."

PK-A01 **SAFE** *Adults*

A home cook follows a tested pickling recipe using white distilled vinegar with 5% acidity, adding fresh dill and garlic to cucumbers. They process the jars in a boiling water bath for the recommended time at their altitude and store the sealed jars in a cool, dark place.

Why: Using vinegar with 5% acidity ensures the proper acidity level to prevent botulism and other pathogens. Following a tested recipe and processing times adjusted for altitude ensures the pickles reach a temperature that kills harmful organisms. Fresh spices enhance flavor without impacting safety, and proper water bath canning seals the jars, preventing contamination.

Volunteer notes: OSU/USDA standards require using vinegar with 5% acidity and processing jars according to altitude-adjusted times in a boiling water bath or pressure canner. Following tested recipes prevents underprocessing. Fresh ingredients and correct headspace are critical to safety.

Source: NCHFP Pickled eggs - "Use vinegar with 5% acidity...Do not use homemade vinegar...Follow instructions for processing times at altitudes...Use fresh spices...White distilled vinegar has a sharp pungent acid taste"

PK-A02 **NOT SAFE** *Adults*

Someone attempts to pickle fresh cucumbers using homemade vinegar of unknown acidity, skipping processing in a water bath or pressure canner and storing the jars at room temperature unsealed.

Why: Homemade vinegar can vary widely in acidity, which makes it impossible to guarantee food safety. Skipping processing prevents killing harmful bacteria or sealing out new contamination. Storing unsealed jars at room temperature allows growth of pathogens like Clostridium botulinum, risking botulism.

Volunteer notes: Using vinegar with unknown acidity or homemade vinegar is unsafe for pickling. Processing jars (water bath or pressure canning) seals and heats jars to destroy bacteria. Unsealed jars and improper acid levels risk botulism and spoilage.

Source: PNW 355 Pickling vegetables - "Do not use homemade vinegar. It varies in acidity...process jars for recommended time...vinegar is the most important ingredient in quick-pickle recipes"

PK-A03 **IT DEPENDS** *Adults*

A gardener picks cucumbers and soaks them in a lime-water solution to firm them before pickling. They then rinse and soak the cucumbers in fresh water twice, as directed, but are unsure if the rinsing was sufficient to remove all the lime before pickling.

Why: Soaking cucumbers in pickling lime can help maintain firmness but it is essential to remove all excess lime by multiple rinses and soakings. If lime residue remains, it can raise the pH, making the pickles unsafe by allowing harmful bacteria to grow. Safety depends on thorough rinsing and following recipe directions exactly.

Volunteer notes: If using lime for firmness, excess lime must be removed by multiple rinses and soaks. Improper rinsing and leftover lime can raise pH above safe levels, risking botulism and spoilage. Confirm thorough rinsing for safe pickles.

Source: PNW 355 Pickling vegetables - "Cucumber or vegetable slices are sometimes soaked in a lime and water solution for 12 to 24 hours before pickling. Be sure to remove excess lime by rinsing and soaking the cucumbers for 1 hour in fresh, cold water and then repeating the process two more times. The excess lime must be removed."

JAMS & JELLIES

JJ-K01 **SAFE** *Kids (under 10)*

You see a jar of homemade strawberry jam in the fridge. The lid is sealed tight, and the jam looks shiny and smooth with no fuzzy spots or strange smells.

Why: When jam jars have tight lids and the jam looks clean and fresh, it is safe because there are no signs of mold or spoilage. Keeping jam in the fridge after opening helps stop germs from growing. When jam is made and stored well, it's a yummy and safe treat.

Volunteer notes: Properly sealed jams with no mold and stored in the refrigerator after opening are safe to eat. High acid and sugar content prevent bacteria growth. Check for signs of spoilage before use.

Source: SP 50-765 Low sugar fruit spreads - "Spreads produced using tested recipes and processed properly are safe to store at room temperature; after opening, refrigeration is important"

JJ-K02 **NOT SAFE** *Kids (under 10)*

You see a jar of blueberry jelly sitting on the kitchen counter. The lid is sticking up a little bit, and you notice some fuzzy white stuff growing on top of the jelly.

Why: When the jar lid is popped up, it means air can get inside, and that lets mold grow. Mold on jelly can make people feel sick. Even if just a little mold shows, it can be all through the jelly. It's safest to throw away jelly that has mold or a lid that is popped up.

Volunteer notes: Broken seals and visible mold on jams/jellies indicate spoilage and potential danger from mold toxins. All contents should be discarded to avoid foodborne illness.

Source: SP 50-934 USDA molds on food - "Foods like jams can have dangerous molds when lids are popped up; mold grows below surface and moldy jams should be discarded"

JJ-K03 **IT DEPENDS** *Kids (under 10)*

Grandma left a jar of grape jelly on the counter overnight. It looks okay, but you smell a little sour smell when you open the jar.

Why: If grape jelly smells sour or funny, it might be starting to spoil because it was left out too long. But sometimes jelly can smell a little different and still be safe if it was sealed well and cooled quickly. It's safest to ask an adult to check for other signs like mold or bubbles before eating.

Volunteer notes: Jam and jelly safety depends on time left unrefrigerated and signs of spoilage such as off-odor, mold, or bubbling. High sugar and acid slow spoilage but do not stop it completely at room temperature once opened.

Source: SP 50-934 USDA molds on food - "Don't leave perishables like opened jams out more than 2 hours; use leftover jams quickly to prevent mold and bacterial growth"

JJ-W01 **SAFE** *Tweens (10-14)*

You watch a grown-up make strawberry jam using a recipe that has just the right amount of fruit, sugar, and a special ingredient called pectin. After cooking the jam, the grown-up carefully pours it into clean jars and seals them tightly while the jam is still warm.

Why: When a grown-up follows a tested recipe and uses clean jars sealed right away, it helps keep the jam safe from going bad. The sugar and pectin help make the jam thick and stop mold or bacteria from growing. Keeping everything clean and sealed helps the jam stay fresh for a long time.

Volunteer notes: Safe jam making depends on using a tested recipe to get the right sugar, acid, and pectin levels and sealing jars while hot to prevent contamination. Tested recipes ensure safety and shelf stability.

Source: SP 50-632 Making berry syrups at home - "The correct proportions of fruit, pectin and sugar are essential for gelling... Follow the preparation instructions in the pectin package. Process low - and no - sugar spreads in a boiling water canner for 10 minutes."

JJ-W02 **NOT SAFE** *Tweens (10-14)*

You find a jar of homemade blackberry jelly in the pantry with fuzzy mold growing on the surface. You decide to scrape off the mold and eat the rest of the jelly.

Why: Mold on jams or jellies means the whole jar could be unsafe, even if you only see mold on top. The mold may have spread below the surface or made harmful toxins. It's best to throw moldy jam away to avoid getting sick.

Volunteer notes: Mold can penetrate moist foods like jams, and scraping mold off doesn't remove all risks. Moldy high-moisture foods should be discarded to prevent illness.

Source: SP 50-632 Making berry syrups at home - "Foods with high moisture content can be contaminated below the surface. Moldy foods may also have bacteria growing along with the mold. The mold could be producing a mycotoxin."

JJ-W03 **IT DEPENDS** *Tweens (10-14)*

You helped make freezer jam with fresh raspberries, but the recipe used less sugar than usual. You put the jam in the freezer right after making it and kept it cold until eaten.

Why: Freezer jams must have enough sugar or acid to prevent bacteria growth. Using less sugar can be safe only if the recipe is made for low-sugar freezer jam and you keep the jam frozen or very cold. If not kept cold or if the recipe isn't tested for safety, it might spoil.

Volunteer notes: Low sugar freezer jams need tested recipes and must be stored frozen or cold to be safe. Sugar or acid helps prevent bacteria from growing.

Source: SP 50-632 Making berry syrups at home - "Because low sugar fruit spreads do not benefit from the thickening and preservative effects of high sugar ratios, home preservers should use these tested techniques to ensure a safe product."

JJ-T01 **SAFE** *Teens (14-18)*

You make strawberry freezer jam using a tested recipe that calls for mashing fresh strawberries with added sugar and then pouring the mixture into clean containers. You store it promptly in the freezer and only take out small amounts to eat within a few weeks.

Why: Freezer jams made with tested recipes and stored frozen promptly stay safe because the cold keeps bacteria from growing. Sugar helps preserve the fruit by lowering water activity, and the recipe acidity ensures safety. Freezer jam is not heat processed, so it must be kept frozen and used soon after opening.

Volunteer notes: According to OSU Extension, uncooked freezer jams are safe when made with tested recipes, proper sugar levels, and stored frozen. They are not heat processed but rely on freezing and sugar for preservation. Once opened, they must be refrigerated and used quickly to prevent spoilage or mold.

Source: SP 50-763 Uncooked freezer jams - "Uncooked freezer jams use sugar and freezing to keep jams safe without heat processing. Use *tested recipes and keep frozen for safety."*

JJ-T02 **NOT SAFE** *Teens (14-18)*

You try to make jelly at home but decide to reduce the sugar way below the recipe's recommendation without using any special low-sugar pectin. You cook the jelly and can it in jars using a water bath canner.

Why: Reducing sugar too much in jam or jelly without using special low-sugar pectin can make the product unsafe. Sugar helps preserve by lowering water activity and helping gel formation. Without enough sugar or special pectin, the jelly may not set properly and can allow bacteria or mold to grow even after canning.

Volunteer notes: OSU guidance says traditional jams and jellies require high sugar amounts for safety and gel formation. Simply cutting sugar drastically or skipping sugar risks runny, unsafe jelly. Use low-methoxyl pectin for low-sugar spreads, and always follow tested recipes to ensure proper acidity and safety.

Source: SP 50-765 Low sugar fruit spreads - "Simply reducing sugar in jam recipes results in runny and potentially unsafe products. Follow tested recipes using special pectin for low-sugar jams."

JJ-T03 **IT DEPENDS** *Teens (14-18)*

You want to make elderberry jam and find a recipe online. The recipe uses American elderberries and calls for a high ratio of sugar to berry pulp, but you consider lowering the sugar amount to make it healthier.

Why: American elderberries can be low acid, meaning safety depends on sugar amount. High sugar ratios reduce water activity to stop botulism bacteria. Lowering sugar in this jam can make it unsafe unless you follow a tested recipe designed for low sugar. You must weigh ingredients exactly and process jars properly or safety is compromised.

Volunteer notes: OSU Extension states American elderberry jams must have a high sugar ratio for safety since these berries can be low acid. Changing sugar amounts without tested guidance risks botulinum toxin. Always follow tested high-sugar elderberry recipes or use only fully acid and sugar balanced recipes.

Source: EM 9446 Playing it safe when preserving elderberries - "American and European elderberries can be low in acid and only safe canned with specific high sugar ratios to prevent Clostridium botulinum."

JJ-A01 **SAFE** *Adults*

You followed a tested recipe to make elderberry jam, weighing berries and sugar exactly, boiling the jam to gel point, then processed the jars in a boiling water bath for 10 minutes at 2,000 feet elevation.

Why: Elderberry jams require precise ratios of sugar to fruit and correct heat processing to prevent bacterial growth. This recipe uses enough sugar to lower water activity and processing time adjusted for altitude ensures safety. Following a tested recipe with these steps prevents foodborne illness.

Volunteer notes: OSU/USDA standards require elderberry jams to have high sugar ratios and processing in boiling water canner with time adjusted by elevation (10 minutes at 2,000 ft). This prevents Clostridium botulinum growth. A common question is, 'Can I reduce sugar in elderberry jam?' Answer: No, not without special low-sugar tested pectin recipes.

Source: EM 9446 Playing it safe when preserving elderberries - "High-sugar elderberry jams must be processed in a boiling water canner, adjusting time for elevation; sugar ratio and weights must be exact to prevent bacterial growth."

JJ-A02 **NOT SAFE** *Adults*

After making peach jam with a traditional recipe, you canned it in a boiling water bath but skipped adjusting the processing time for your altitude of 5,500 feet, using only the low-altitude (5 minute) time.

Why: Processing time must increase with altitude because water boils at lower temperatures at higher elevations. Using the low-altitude time at 5,500 feet means the jam was underprocessed, risking spoilage and botulism. Always follow altitude-adjusted times to ensure safety.

Volunteer notes: OSU/USDA guidelines specify processing times increase with altitude (e.g., 10 minutes between 1,001-6,000 feet). Underprocessing at altitude risks unsafe food due to inadequate heat penetration. Common question: 'How do I adjust processing time for altitude?' Use reliable extension charts for altitude corrections.

Source: EM 9446 Playing it safe when preserving elderberries - "Process jams for 5 minutes under 1,000 feet, 10 minutes 1,001 to 6,000 feet, 15 minutes above 6,000 feet in a boiling water canner."

JJ-A03 **IT DEPENDS** *Adults*

You made a low-sugar jam using a regular pectin product but decided to reduce the sugar by half without changing the pectin or acid, then processed it normally in a boiling water bath.

Why: Reducing sugar in traditional jam recipes without using low-sugar pectin or altering acid levels can result in unsafe jam because sugar preserves and helps gel formation. Safety depends on using tested recipes designed for low sugar

and following processing instructions precisely.
Volunteer notes: Reducing sugar in jams requires special low-sugar or no-sugar pectin brands that have tested recipes to ensure safety and gel. Using regular pectin with less sugar can yield runny, unsafe products. Question: 'Can I just reduce sugar in my old recipe?' Answer: No; use a low-sugar pectin and follow its recipe for safety.
Source: SP 50-632 Making berry syrups at home - "Simply reducing sugar in a traditional jam or jelly recipe will result in a runny and potentially unsafe product; use low-methoxyl pectin with tested recipes."

FREEZING & COLD STORAGE

FZ-K01 SAFE *Kids (under 10)*

You find a box of frozen peas in the freezer, and the freezer door is closed tight. The peas look bright green and don't smell funny at all. Your family always checks the freezer temperature and keeps it very cold.

Why: When frozen food is kept at the right cold temperature, it stays safe to eat. If the freezer door is closed and the food looks and smells normal, it means the food was stored properly and won't make you sick.
Volunteer notes: Frozen foods kept at 0°F or below remain safe. Foods maintaining ice crystals can be refrozen safely. Proper freezer temperature keeps harmful bacteria from growing .
Source: PNW 214 Freezing fruits and vegetables - "Store them at 0°F (-18°C) or below. Foods lose quality and nutritive value much faster at higher temperatures."

FZ-K02 NOT SAFE *Kids (under 10)*

You see a leftover sandwich on the kitchen counter all day while your family was busy. The sandwich has cold cuts and cheese. Later you find it forgotten there and wonder if it's okay to eat.

Why: Leaving foods with meat or cheese out at room temperature for many hours lets bad germs grow quickly. Eating it could give you a tummy ache. Foods with meat or dairy must be kept cold until eaten.
Volunteer notes: Food left in the danger zone (between 40°F and 140°F) longer than 2 hours can grow harmful bacteria. Always refrigerate perishable foods quickly to avoid foodborne illness .
Source: PNW 296 Freezing convenience foods that you've prepared at home - "Low acid foods ... that have been kept in the danger zone (40 - 140 °F) for more than 2 - 3 hours are a safety concern."

FZ-K03 IT DEPENDS *Kids (under 10)*

You notice some frozen berries in the freezer have frost on the outside and the bag is partially open. They don't smell bad, but the berries look a little mushy when you touch them.

Why: The frost and mushy feel mean the berries might have freezer burn, which can change taste and texture but usually does not make you sick. If they don't smell bad or look moldy, they are safe to eat but might not taste as good.
Volunteer notes: Freezer burn happens when frozen food is exposed to air due to damaged packaging. It affects quality but not safety. Check for bad odors or mold to decide if food is safe to eat .
Source: PNW 214 Freezing fruits and vegetables - "Food quality is affected by freezer burn, but food is safe if kept frozen without obvious spoilage signs."

FZ-W01 SAFE *Tweens (10-14)*

You watched your family wash and cut fresh green beans to freeze. They boiled the beans for a short time to keep them tasty and then packed them into special freezer bags, squeezing out the air before putting them in the freezer at 0°F. This helps the green beans stay fresh and safe to eat later.

Why: Blanching green beans by briefly boiling them before freezing helps keep them fresh and good to eat. Packing the beans in airtight bags and freezing at the right temperature stops bacteria from growing, so the beans can stay safe for several months.
Volunteer notes: Freezing vegetables after blanching and storing at 0°F or below maintains quality and safety for many months. Proper packing and quick freezing reduce spoilage and bacteria growth .
Source: PNW 214 Freezing fruits and vegetables - "Store them at 0°F (-18°C) or below. Most fruits and vegetables maintain high quality for 8 to 12 months."

FZ-W02 NOT SAFE *Tweens (10-14)*

Your friend left a frozen pizza out on the kitchen counter all afternoon to thaw. Later, your friend put it back in the freezer without cooking it.

Why: Leaving frozen or refrigerated foods at room temperature too long lets harmful bacteria multiply quickly. Refreezing thawed food without cooking it first can keep bacteria alive and might cause food poisoning.
Volunteer notes: Thawed perishable food should be cooked before refreezing to kill bacteria. Foods left above 40°F for over 2 hours are unsafe to eat or refreeze. This is a common freezing and cold storage mistake.
Source: PNW 214 Freezing fruits and vegetables - "If the temperature has warmed above 40°F (5°C), foods may not be safe for refreezing."

FZ-W03 IT DEPENDS *Tweens (10-14)*

You see a glass jar of homemade soup in the freezer, but the jar's lid isn't closed tightly. The soup looks frozen and really cold. Is it okay to eat the soup?

Why: It depends. If the soup is still frozen or very cold, it's usually safe to keep it by putting it in the fridge or freezer again. But if the soup has warmed up to room temperature, it's safer to throw it away to avoid getting sick.
Volunteer notes: Safety of frozen canned or jarred food depends on whether the seal is intact and the food temperature. Broken seal with cold or frozen food may be safe if handled properly; room temperature makes it unsafe. Boiling low acid foods before use adds safety margin.
Source: PNW 296 Freezing convenience foods that you've prepared at home - "If the seal has broken and the food is still frozen or cold (refrigerator temperature , 40 ° F or below), it may be safely salvaged."

FZ-T01 SAFE *Teens (14-18)*

You froze a batch of homemade vegetable soup in clean, airtight containers. The freezer temperature is kept steady at 0°F (-18°C) or below. You always check the freezer thermometer regularly.

Why: Keeping frozen foods at 0°F or colder stops bacteria growth and preserves quality. Using airtight containers prevents freezer burn and contamination. Regularly checking the freezer thermometer helps ensure safe storage conditions.
Volunteer notes: Store frozen foods at 0°F (-18°C) or below. Using proper containers and maintaining steady freezer temperatures preserves safety and quality. Freezer thermometers help monitor conditions.
Source: PNW 214 Freezing fruits and vegetables - "Store them at 0°F (-18°C) or below. Foods lose quality and nutritive value much faster at higher temperatures. Use a freezer thermometer to monitor the temperature."

FZ-T02 NOT SAFE *Teens (14-18)*

You left a stuffed turkey breast in the freezer, but the power went out for about 3 days. You opened the freezer a few times during that time to check the food. Some turkey looks partially thawed and soft.

Why: If freezer temperature rises above 40°F for long periods, bacteria can grow even if food looks okay. Opening the freezer lets cold air out, speeding thawing. Partially thawed poultry can be unsafe to eat and should be discarded.
Volunteer notes: Food in a freezer without power more than 2 days may thaw and become unsafe. Opening the freezer accelerates thawing. Thawed poultry showing soft texture or off odors should be discarded to avoid illness.
Source: PNW 214 Freezing fruits and vegetables - "A full load of food will stay frozen for up to 2 days if the freezer is not opened. If the temperature has warmed above 40°F (5°C), foods may not be safe for refreezing. Do not eat thawed vegetables that are above 40°F. Unsafety may not show obvious signs."

FZ-T03 IT DEPENDS *Teens (14-18)*

You froze fresh fish you caught the same day, but you didn't ice the fish immediately-you froze it right after cleaning. Now you wonder if it's safe to eat after freezing.

Why: Freezing fish soon after catch preserves safety, but the timing matters. Fish that were iced immediately after bleeding pass rigor mortis slowly, which maintains better quality and safety. Fish frozen before rigor or without icing can have tough texture and lower quality. If the fish wasn't fresh or iced properly before freezing, there might be safety risks.
Volunteer notes: Fresh fish should be bled and iced immediately after catch and before freezing to ensure safety and quality. Freezing slows bacterial growth but won't fix poor initial handling. If fish handling before freezing was poor, discard it.
Source: PNW 586 Home freezing seafood - "Fish should be bled as soon as they are caught, and then iced immediately. If you're ever in doubt as to the freshness of seafood, don't freeze it. Freezing can only slow the loss of quality-it can't put any quality back."

FZ-A01 SAFE *Adults*

You froze a batch of cooked chili in shallow containers, leaving about 1-inch headspace, and your freezer is kept at 0°F (-18°C). The chili was cooled in an ice water bath before freezing. You label and date the containers before placing them in the freezer.

Why: Freezing cooked food in shallow containers with appropriate headspace allows for rapid cooling and prevents unsafe temperature ranges that promote bacteria growth. Keeping the freezer at 0°F (-18°C) or below maintains food safety and quality. Labeling and dating packages helps rotate and use food within recommended times to maintain quality.

Volunteer notes: According to OSU and USDA standards, rapid cooling before freezing and maintaining freezer temperatures at or below 0°F keeps food safe. Appropriate headspace prevents container bursting and enables even freezing. Labeling assists in quality management. A follow-up: 'Why is headspace important?' helps learners understand expansion during freezing.

Source: PNW 296 Freezing convenience foods that you've prepared at home - "Use about 1-inch headspace in containers; store frozen foods at 0°F or below; use a freezer thermometer to monitor temperature; label and date packages to rotate supply"

FZ-A02 NOT SAFE *Adults*

You stored home-canned green beans in your garage during winter, where temperatures dropped below 20°F. Some jars cracked, and others froze but did not crack. Later, you thawed and consumed the food from the cracked jars without reheating.

Why: Freezing can cause jars to crack or seals to break, allowing contamination. Consuming food from cracked jars without reheating can cause foodborne illness. Canned goods should not freeze; if they do, evaluate jars carefully and boil low-acid foods for 10 minutes before use to ensure safety. If jars are cracked or seals broken and food has thawed at room temperature, discard immediately.

Volunteer notes: Storing canned foods below 34°F can cause freezing damage leading to unsafe food. Broken seals or cracked jars compromise safety and require disposal or boiling low-acid foods for safety. A typical Q&A: 'How can you prevent freezing damage in canned foods?'

Source: PNW 296 Freezing convenience foods that you've prepared at home - "If jars cracked or seals broken after freezing and thawing, discard the food. Boil low acid foods 10 minutes for extra safety if sealed jars thawed but seal intact."

FZ-A03 IT DEPENDS *Adults*

You froze a Seafood stew containing fish and shellfish in a large, deep container without dividing into smaller portions. The freezer temperature is usually 0°F, but sometimes rises above 5°F when the door is opened frequently. You wonder if this stew is still safe to eat.

Why: Seafood freezes safely at 0°F, but large, deep containers freeze more slowly, allowing bacteria growth during prolonged cooling. Frequent temperature fluctuations above 5°F may compromise safety. If the stew was cooled quickly to 0°F or below and has remained frozen, it is safe. If thawing occurred or temperatures rose above 40°F for significant time, safety is compromised. Always consider how quickly food was frozen and the freezer temperature history.

Volunteer notes: Freezing safety depends on how quickly food is cooled, container size, and freezer temperature consistency. Large containers slow freezing; thaw/freeze cycles can increase risk. Q&A follow-up: 'How do you ensure rapid freezing to keep seafood safe?'

Source: PNW 586 Home freezing seafood - "Freeze seafood rapidly in small portions; large containers delay freezing and may allow bacteria growth. Keep freezer at 0°F or below for safety."

DEHYDRATING & SMOKING

DS-K01 SAFE *Kids (under 10)*

You see some meat jerky drying in the oven, and an adult checks the temperature to make sure it gets really hot before drying it. The jerky looks dry and ready.

Why: Jerky is safe to eat when the meat is heated to the right temperature before or right after drying. This kills germs that can make you sick. Drying the meat until it is dry helps keep it safe to store without spoiling.

Volunteer notes: Jerky must be heated to 160°F (beef) or 165°F (poultry) before or immediately after drying to destroy harmful bacteria like Salmonella and E. coli. Proper drying ensures shelf-stable jerky for safe storage at room temperature for 1-2 months. This step is critical for homemade jerky safety.

Source: PNW 632 Making jerky at home safely - "Jerky can be considered "done" and safe to eat only when meat has been heated sufficiently to 160°F and poultry to 165°F before or immediately after the dehydration process. This destroys any pathogens present and creates jerky that is dry enough to be shelf-stable."

DS-K02 **NOT SAFE** *Kids (under 10)*

You notice some fish in the smoker that looks wet and smoky, but it hasn't reached the right temperature all the way through. The adult leaves it there.

Why: Fish has to be cooked or smoked until it is hot enough all the way through to kill germs that can make people sick. Smoking fish without reaching that safe temperature can let dangerous germs stay alive. Eating fish that isn't cooked enough is not safe.

Volunteer notes: Smoking fish at unsafe temperatures can allow pathogens to survive, causing foodborne illness. Fish must reach safe internal temperatures to eliminate bacteria and parasites. Undercooked or improperly smoked fish poses a serious safety risk.

Source: PNW 238 Smoking fish at home - "Fish must be smoked or cooked to a safe temperature to destroy pathogens; insufficient heating or smoking can lead to health risks."

DS-K03 **IT DEPENDS** *Kids (under 10)*

You see an adult drying fruit on the counter near the open window. Sometimes flies come inside and land on the fruit. The fruit looks okay, but you wonder if it's safe.

Why: Drying fruit is safe if it stays clean and dry, but if insects land on it or it gets dirty, it might have germs that can cause sickness. It depends on whether the fruit stays protected from bugs and dirt and dries well. If you see any mold, bad smell, or bugs, it is not safe.

Volunteer notes: Dehydrated fruits must be protected from insects and contamination during drying to remain safe. If fruit is exposed to flies or dirt, it could be contaminated. Safety depends on cleanliness, drying completeness, and absence of mold or spoilage signs.

Source: PNW 397 Drying fruits and vegetables - "Proper drying requires protection from insects and dirt; contamination during drying affects safety, and visible mold or spoilage indicates the product is unsafe."

DS-W01 **SAFE** *Tweens (10-14)*

You helped a grown-up dry apple slices in the dehydrator. They made sure the apples were cut evenly and dried at the right temperature until there was no moisture left. Then, the dried apples were stored in an airtight container in a cool, dark place.

Why: This is safe because drying removes moisture that bacteria need to grow. Keeping dried fruit in a sealed container away from light and heat keeps it fresh and safe to eat. It's important that the drying temperature was right and the fruit was fully dried to stop germs from growing.

Volunteer notes: Dried fruits are safe when dried to proper moisture levels and stored airtight in a cool, dark place to prevent spoilage and mold. This follows OSU Extension and USDA guidelines on drying fruits safely.

Source: PNW 397 Drying fruits and vegetables - "Dried fruits, nuts, vegetables, and jerky must be dried thoroughly and stored in airtight containers in a cool, dark place to keep them safe and high quality."

DS-W02 **NOT SAFE** *Tweens (10-14)*

A friend bought meat jerky from a store that said 'homemade' but it was not heated to at least 160°F on the meat. They left the jerky out on the counter for a few days before eating it.

Why: Jerky must be heated enough to kill harmful bacteria-usually to 160°F. If it isn't heated properly, the jerky can still have dangerous germs. Also, jerky left out too long without proper drying or heating can spoil and cause sickness.

Volunteer notes: Home-prepared jerky must be heated to 160°F (for beef) to destroy pathogens such as Salmonella and E. coli. Improper heating and storage increases risk of foodborne illness. Follow safe heating protocols for jerky.

Source: PNW 632 Making jerky at home safely - "Jerky is safe only when meat has been heated sufficiently to 160°F before or immediately after dehydration to destroy pathogens and ensure shelf stability."

DS-W03 **IT DEPENDS** *Tweens (10-14)*

You helped a grown-up prepare fish by putting it in the smoker. They didn't watch the smoker to make sure it stayed warm the whole time. Sometimes, the smoker got cooler than it should for a while.

Why: Smoking fish can be safe when the smoker stays warm enough to keep germs from growing. If the smoker gets too cool for some time, germs might grow and make the fish unsafe to eat. So, it depends on how long the smoker was too cool and if the grown-up made sure the fish was cooked well after.

Volunteer notes: Cold-smoking or smoking at too low temperatures can allow harmful bacteria to survive. Maintaining proper temperature above 140°F during smoking is crucial for safety. If temperatures drop, safety depends on time and later cooking steps.

Source: PNW 238 Smoking fish at home - "Fish must be smoked at safe temperatures, usually above 140°F, to kill pathogens. If temperatures fall below this, safety depends on time and subsequent cooking or handling."

DS-T01 **SAFE** *Teens (14-18)*

You prepare beef jerky by slicing lean meat, marinate it in the refrigerator for 8 hours, then dehydrate it until dry. After drying, you heat the jerky in an oven at 275°F for 10 minutes before storing it at room temperature.

Why: Heating the jerky after drying to 160°F or higher kills harmful bacteria that might survive the drying process. This step ensures the jerky is safe to eat and can be stored at room temperature for 1-2 months without risk.

Volunteer notes: According to USDA and OSU Extension, safe jerky must be cooked to 160°F (beef) or 165°F (poultry), either before or after drying. Post-drying heating at 275°F for 10 minutes is an effective method to ensure safety. This eliminates pathogens like Salmonella and E. coli, preventing foodborne illness.

Source: PNW 632 Making jerky at home safely - "Researchers found heating jerky in a 275°F oven for 10 minutes after drying ensures it is safe and shelf-stable for 1-2 months at room temperature."

DS-T02 **NOT SAFE** *Teens (14-18)*

You hot-smoke fish by placing it in a smoker at 90°F, keeping it there for several hours without adding salt or cooking it to a higher temperature, then storing it at room temperature.

Why: Smoking fish at low temperatures like 90°F without sufficient salt or cooking doesn't kill dangerous bacteria like Clostridium botulinum. Without proper heating and salting, storing smoked fish at room temperature can cause lethal food poisoning.

Volunteer notes: Safe hot-smoking requires heating fish to at least 150°F (preferably 160°F) and salting it to 3.5% water phase salt throughout. Without these, bacteria can grow and produce toxins. Refrigeration at 38°F or less is essential if these conditions are not met.

Source: PNW 238 Smoking fish at home - "Fish must be heated to 150-160°F and salted adequately during smoking to prevent dangerous bacterial growth; cold smoking or low heat is unsafe without refrigeration."

DS-T03 **IT DEPENDS** *Teens (14-18)*

You dry sliced apricots in a dehydrator. You wonder if it's safe to eat them without treating them first since apricots are low-acid fruits. You consider just drying them or pretreating with lemon juice or another acid.

Why: Drying low-acid fruits like apricots can be safe if you pretreat them with acid (like lemon juice) or use a tested drying method because low-acid fruits can spoil or grow bacteria. Without pretreatment, the safety depends on drying time, temperature, and handling.

Volunteer notes: Low-acid fruits need pretreatment (acid dip or sulfuring) to prevent spoilage and pathogen growth. Proper drying reduces moisture to safe levels. Without pretreatment, drying alone might not make them shelf-stable or safe, so it depends on the exact method used.

Source: PNW 397 Drying fruits and vegetables - "Low-acid fruits require acid pretreatment to prevent spoilage and ensure safety during drying and storage; untreated drying safety depends on method and conditions"

DS-A01 **SAFE** *Adults*

Jamie uses a food dehydrator with a thermostatically controlled temperature dial set to 155°F and dries thin strips of beef for jerky. Jamie ensures the temperature is maintained steadily and follows a tested jerky recipe.

Why: Maintaining a drying temperature of at least 145°F, preferably around 155°F, and following a tested recipe ensures that harmful bacteria are killed during jerky preparation. Using a dehydrator with a thermostat and verifying the actual temperature helps guarantee safety. This method aligns with OSU Extension recommendations for safe jerky drying.

Volunteer notes: Safe jerky drying requires maintaining temperatures 145°F-155°F throughout drying and using recipes vetted for safety. Temperatures lower than this may allow bacterial survival. Confirm your dehydrator's temperature with an accurate thermometer before use.

Source: PNW 632 Making jerky at home safely - "Use a thermometer to determine that the empty, operating dryer can maintain a temperature of at least 145°F. Do not use dehydrators with factory preset temperatures that cannot be controlled."

DS-A02 **NOT SAFE** *Adults*

Alex smokes fish at home using a traditional cold-smoking method without cooking the fish to a safe temperature or using a tested smoker design, then stores the fish at room temperature.

Why: Cold smoking does not cook fish and does not reliably kill harmful bacteria or parasites. Without further cooking or

preservation steps like freezing or curing, storing cold-smoked fish at room temperature risks foodborne illness. Proper hot-smoking or following tested preservation methods is necessary for safety.
Volunteer notes: Cold smoking alone does not make fish safe; it does not reach temperatures to destroy pathogens. Foods not cooked or cured and stored at unsafe temperatures can cause illness. Always follow up-to-date, tested recipes.
Source: PNW 238 Smoking fish at home - "Cold smoking does not cook fish or kill harmful bacteria; fish must be hot smoked or frozen to ensure safety."

DS-A03 **IT DEPENDS** *Adults*

Morgan dries sliced apples using a home dehydrator, but isn't sure if their home is at 3,000 feet elevation where drying times might need adjustment. Morgan follows a standard drying temperature of 135°F.

Why: Drying times and temperatures sometimes need adjustment based on altitude (typically above 1,000 feet) due to lower air pressure affecting water evaporation. At higher elevations like 3,000 feet, longer drying times or slightly higher temperatures may be needed to ensure food safety and prevent spoilage. Morgan should consult altitude-specific guidelines to be sure.
Volunteer notes: Altitude affects drying efficiency. Above 1,000 feet, drying times usually increase. For high acid dried foods like fruit, slight temperature adjustments help ensure moisture removal and inhibit microbial growth.
Source: PNW 397 Drying fruits and vegetables - "Drying times and oven settings may need to be adjusted for altitude where air pressure is lower and moisture evaporates differently."